

Imaging M87* with a Deep Image Prior

Student instructions and exercise guide (S)

EHT interferometric imaging practical – Colab/CUDA

Purpose

Reconstruct a 64×64 image of M87* from sparse Event Horizon Telescope (EHT) visibility amplitudes and closure phases. You will use a Deep Image Prior (DIP) neural network, assess the result in observable space, and test how scientific conclusions depend on resolution, explicit regularization, robustness choices, closure phases, and initialization.

Notebook used for this handout: `EHT_DIP_Colab_Tutorial.ipynb`. The notebook is the executable reference; this handout tells you what to run, what to change, and what evidence to record.

Learning outcomes

By the end of the practical you should be able to:

1. inspect and understand interferometric coverage in the (u, v) plane;
2. relate image pixels to complex visibilities through the Fourier measurement equation;
3. explain how amplitudes, closure phases, a neural parameterization, and explicit penalties constrain the inverse problem;
4. evaluate a reconstruction in data space rather than by visual appearance alone; and
5. distinguish measurement-supported structure from prior-sensitive structure.

1 Quick-start instructions

1.1 Start a CUDA runtime

Open the notebook in Google Colab. Select **Runtime** → **Change runtime type** → **T4 GPU** (or another CUDA GPU), then run the cells in order.

Have these files ready:

- `eht_dip.py` – reconstruction model, optimizer, and bootstrap routine;
- `SR1_M87_2017_095_lo_hops_netcal_StokesI.npz` – the supplied M87* observables.

If they are not already under `/content`, the notebook opens an upload dialog. Do not rename either file. The environment check must print a PyTorch version, GPU name, and CUDA runtime. If CUDA is unavailable, stop and change the runtime type before continuing.

1.2 Inspect the data before imaging

Run the loading and plotting cells. Confirm that all arrays are finite, all quoted uncertainties are positive, and both amplitude and closure-phase arrays are present. Record:

- the numbers of visibility amplitudes and closure phases;
- the maximum baseline length in $G\lambda$;
- the amplitude range in Jy;
- gaps, preferred directions, and maximum radius in the (u, v) coverage.

For a small field of view, the forward model is

$$V(u, v) = \iint I(l, m) \exp[-2\pi i(ul + vm)] dl dm. \quad (1)$$

Visibility amplitudes constrain $|V|$ and the flux scale. Closure phases constrain

$$\psi = \arg(V_1 V_2 V_3), \quad (2)$$

while canceling station-based phase errors around a closed triangle. Neither observable fixes the absolute image position.

1.3 Run the reference reconstruction

The DIP parameterizes the positive image as

$$I_{\theta} = \text{softplus}[f_{\theta}(z)], \quad (3)$$

where the input z is fixed and only the network parameters θ are optimized for this data set. The network architecture is an implicit prior; TSV and compactness are explicit priors.

Use the notebook’s current reference settings:

Setting	Reference value	Interpretation
<code>NPIX</code>	64	Image width and height; must be a positive multiple of 32
<code>PIXEL_SIZE_UAS</code>	2.0	Pixel size in μas ; field of view is $128 \mu\text{as}$
<code>NUM_ITER</code>	4,000	Optimization budget for the reference run
<code>amp_weight, cp_weight</code>	1.5, 1.0	Relative weights of amplitude and closure-phase data terms
<code>lambda_tsv</code>	1.0	Squared spatial-gradient penalty
<code>lambda_compact</code>	0.5	Penalty on emission outside the compact region
<code>r_compact_uas</code>	40	Compact-region radius in μas
<code>t_nu_amp, t_nu_cp</code>	3, 3	Degrees of freedom of the robust Student- t losses
<code>gauss_fwhm_uas</code>	20	Angular scale of the Gaussian-plus-noise DIP input
<code>blur_sigma0_pix</code>	2.0	Initial coarse-to-fine output blur; it decays to zero

The objective is schematically

$$\mathcal{L} = w_A \mathcal{L}_{t,A} + w_\psi \mathcal{L}_{t,\psi} + \lambda_{\text{TSV}} R_{\text{TSV}} + \lambda_{\text{comp}} R_{\text{comp}} + \dots \quad (4)$$

The Student- t terms reduce the influence of extreme residuals. Closure-phase differences are wrapped to $[-\pi, \pi]$.

Use the 400-step quick mode to scan parameter values. Re-run the reference and the most informative variants for 4,000 steps. Change one assumption at a time, keep a run log, and never select a setting from image appearance alone.

1.4 Evaluate and save

Run the data-space diagnostics. Inspect the image, amplitude fit, closure-phase fit, and normalized-residual histograms together. Look for coherent trends with baseline length. The displayed mean χ^2 values use the quoted errors as a diagnostic; they are not the Student- t objective and are not formal reduced χ^2 values without an effective degrees-of-freedom calculation.

Run the export cell and download `eht_dip_results.zip`. It contains the image in NumPy and FITS formats, modeled observables, and the diagnostic plot. Colab storage is temporary.

1.5 Run the weighted bootstrap

The updated notebook uses ten Poisson-weighted realizations with 4,000 optimization steps per realization. Run this only after checking the reference fit. Save the bootstrap mean, pixelwise standard deviation, and image stack. The ensemble probes sensitivity to data reweighting; it does not include all calibration, scattering, or model systematics.

1.6 Record the reference run

Complete this table before beginning the parameter studies:

Quantity	Result
GPU model	
Elapsed time	
Mean amplitude χ^2 diagnostic	
Mean closure-phase χ^2 diagnostic	
Image sum / zero-baseline flux proxy	

Before starting the parameter studies, save the reference outputs under an unambiguous name, decide which comparisons use quick scans and which require 4,000-step runs, adopt common image orientation and color scaling, and record the exact parameter values and seed before every run.

Exercise 1: Resolution and sampling

Question. Estimate the nominal fringe spacing from the longest baseline and compare it with the adopted $2\text{-}\mu\text{as}$ pixel size.

Procedure.

1. Read B_{max} from the data-inspection output. The (u, v) coordinates are in wavelengths.
2. Compute $\theta_{\text{nominal}} \simeq 1/B_{\text{max}}$ in radians.
3. Convert to microarcseconds using $1 \text{ rad} = 206265 \times 10^6 \mu\text{as}$.
4. Calculate how many $2\text{-}\mu\text{as}$ pixels span this angular scale.
5. Compare the result with the $128 \mu\text{as}$ field of view and with the size of structures in the reconstruction.

Interpretation. Explain why pixel spacing is a sampling choice rather than a statement that neighboring pixels contain independent information. Use the directional gaps in the (u, v) coverage to discuss why angular resolution may be anisotropic.

Exercise 2: Implicit and explicit smoothness priors

Question. Which structures are controlled by the convolutional DIP architecture, and which are controlled by total squared variation (TSV)?

Procedure.

1. Hold the data weights, compactness settings, input, image grid, and seed fixed.
2. Explore `lambda_tsv` beginning at 0.5 and increasing toward 10^6 . Include the reference value 1.0. Use short runs for the sweep.
3. Select at least three informative settings (weak, intermediate, strong) and run them long enough to compare convergence fairly.
4. For every setting, compare image morphology, amplitude residuals, closure-phase residuals, and the total image sum.

Interpretation. Identify when increasing TSV suppresses small-scale structure and when it begins to force coherent misfit in data space. Any structure that persists when TSV is weak may still arise from the network's implicit prior; persistence alone does not prove that the data require it.

Exercise 3: Compactness prior and support assumptions

Question. How restrictive can the assumed source support become before the observables reject it?

Procedure.

1. Return to the reference TSV setting.
2. Vary `r_compact_uas` and `lambda_compact`, changing one at a time around the reference values $40\ \mu\text{as}$ and 0.5.
3. Include one deliberately weak constraint and one deliberately restrictive constraint.
4. Inspect emission outside the chosen radius and search for systematic residuals as a function of baseline length.

Interpretation. Distinguish “the reconstruction is compact because the data prefer it” from “the reconstruction is compact because the penalty forbids alternatives.” State the threshold at which the support assumption becomes inconsistent with the data or produces obvious boundary artifacts.

Exercise 4: Robust Student- t versus Gaussian residuals

Question. Which measurements and image features are most affected by the assumed residual distribution?

Procedure.

1. Run the reference robust model with `use_t_amp=True` and `use_t_cp=True`.
2. Run the Gaussian normalized-squared-residual model with both flags set to `False`. Keep all other settings identical.
3. Rank the largest absolute normalized residuals for amplitudes and closure phases. Record their baseline or triangle scales.
4. Compare the residual histograms, coherent trends, morphology, and flux proxy.

Interpretation. Explain how a heavy-tailed likelihood can reduce the leverage of outlying data. Decide whether differences are driven by a few measurements or by a broad mismatch. Do not label a datum “bad” merely because one model gives it a large residual.

Exercise 5: Information carried by closure phases

Question. What image degeneracies become visible when closure phases are removed, even if amplitude residuals remain acceptable?

Procedure.

1. Run the reference reconstruction with `cp_weight=1.0`.
2. Repeat with `cp_weight=0`, keeping the same amplitude model and priors.
3. Compare the amplitude fit first. Then forward-model and display the closure phases of both images even though closure phases were not optimized in the second run.
4. Examine handedness, azimuthal asymmetry, orientation, translation, and any mirror-related ambiguity.

Interpretation. Describe which features visibility amplitudes can constrain and which require phase information. Remember that closure phase is translation invariant, so restoring it does not determine the absolute centroid.

Exercise 6: Initialization, reproducibility, and bootstrap variation

Question. How much image variability comes from non-convex optimization, and how does it compare with variation induced by data reweighting?

Procedure.

1. Locate `ReconConfig.seed` in `eht_dip.py`; the supplied module uses seed 42 internally. Make seed configurable or change it deliberately.
2. Run several otherwise identical reconstructions with different seeds. Use the same iteration count and record every seed.
3. Run the notebook's Poisson-weighted bootstrap and retain the mean, standard-deviation map, and individual images.
4. Align images if your comparison metric is sensitive to the expected translation ambiguity. State explicitly whether and how alignment was done.
5. Compare pixelwise variability, integrated flux, radial profiles, and at least one morphology statistic of your choice.

Interpretation. Seed-to-seed scatter diagnoses optimization and implicit-prior sensitivity; bootstrap scatter diagnoses sensitivity to weighted versions of the measurements. Neither automatically captures calibration or model systematics. Explain which source of variation dominates in your tests.

2 Final questions to think about!

Answer these questions in your conclusion:

1. Which qualitative structures are stable across the acceptable settings?
2. Which structures change with TSV, compactness, likelihood, phase data, or seed?
3. Is there any parameter choice that looks visually attractive but is contradicted by coherent data-space residuals?
4. What additional checks would be needed before making an astrophysical claim?

Troubleshooting

Symptom	Action
CUDA unavailable	Enable a GPU runtime and rerun from the first cell.
File not found	Upload <code>eht_dip.py</code> and the supplied NPZ file without renaming them.
CUDA out of memory	Restart the runtime, retain <code>NPIX=64</code> , and avoid keeping unnecessary image ensembles in GPU memory.
NaN or unstable fit	Restore the reference settings, then change one parameter at a time; check learning rate and regularizer scaling.
Unfinished quick image	Switch off quick mode and compare a 4,000-step run using data-space diagnostics.
Missing output after session	Colab storage is ephemeral; download <code>eht_dip_results.zip</code> before ending the session.

This notebook has been prepared by Joel Sanchez-Bermudez for the 2026 Severo Ochoa school on Advanced Neural Networks organized by the IAA-CSIC / The code provided for the exercises is proprietary and, in case of use for scientific publication, acknowledgments should be provided to the author.